Torsional friction in planar contact

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November 10, 2026

Abstract

Tactile sensors are widely used to detect slip, but much less is known about what can actually be inferred about frictional contact limits from tactile measurements. This question matters because detecting that slip is occurring or imminent is not the same as knowing which tangential and torsional loads the contact can still support. That distinction is important for control design. In this report, we study this question for a single planar contact patch under quasi-static Coulomb friction. We first show that coarse normal force information, consisting of the total normal force and center of pressure, is generally insufficient to determine the admissible tangential wrench set: two contacts can have the same total normal force and CoP while supporting different combinations of tangential force and torsional moment. We then study what can still be inferred from partial measurements. In particular, we derive upper and lower bounds on the admissible wrench set from coarse measurements and from spatially distributed taxel-wise normal forces. The latter provide tighter bounds by reducing the ambiguity on how the normal force is distributed over the contact patch. Taking the torques about a fixed point makes every bound a generalized moment problem: the bounds from taxel forces have closed forms, those combining taxel forces and CoP are an SOCP and an LP, and the gap between them halves, to first order, with each dyadic refinement of the taxel grid. Along a given load, the guaranteed and the possible wrench sets give a certified and an upper load factor, each a convex program, which decide whether the load is surely carried, surely not carried, or undecided by the measurement.

1 Introduction

Modern tactile sensors provide rich local information about contact pressure, shape, and deformation. Yet tactile features used for control are often selected mainly from empirical performance, for example contact pose, resultant force, image features, or local pressure measurements. A more basic question is less often made explicit: given a particular tactile measurement, what can actually be inferred about the physical interaction at the contact? We study this question in the setting of a rigid planar contact patch under Coulomb friction. Rather than trying to characterize every possible tactile signal or manipulation task, we focus on one concrete control-relevant quantity: the set of tangential forces and torsional moments that the contact can support without violating the friction model. The key difficulty is that this admissible wrench set depends on how the normal load is distributed over the patch. Coarse measurements such as the total normal force and center of pressure constrain this load distribution, but do not determine it uniquely. Spatial tactile measurements provide additional information and therefore reduce this ambiguity. The objective of this report is to make this relationship explicit. We ask **how the admissible frictional wrench set can be bounded from different levels of normal force information**, from total force and CoP to spatially distributed taxel-wise forces, **without requiring the full normal pressure field**.

2 Preliminaries (measures)

We briefly introduce the measure-theoretic notation used in the following. Only the notions needed for the contact model are recalled here; standard reference is [1].

2.1 Force as a measure.

Let $P \subset \mathbb{R}^2$ denote the contact patch, equipped with its Borel σ-algebra $\mathcal{B}(P)$. A *nonnegative measure* $\nu_N : \mathcal{B}(P) \to [0, +\infty]$ assigns to every measurable subset $A \subseteq P$ a nonnegative scalar $\nu_N(A)$, and is countably additive, i.e. for any pairwise disjoint measurable sets $A_i \subseteq P$,

$$\nu_N\left(\bigcup_i A_i\right) = \sum_i \nu_N(A_i). \tag{1}$$

In the present context, ν_N represents the (spatially distributed) normal force, i.e. $\nu_N(A)$ is the normal force carried by the region A. Hence $\nu_N(A)$ has units of force (N), and the resultant normal force is $f_N = \nu_N(P)$. We restrict attention to *finite nonnegative Radon measures*, denoted $\nu_N \in \mathcal{M}_+(P)$. Informally, a Radon measure is a "well-behaved" measure. In particular, on the compact sets considered here, it is finite and its value on measurable sets can be approximated from compact subsets. For our purpose, it essentially means that our measure does not assign an infinite force to P and it can represent either the force induced by an ordinary pressure field or an idealized concentrated point/line load.

We represent the tangential force by an $\mathbb{R}^2$-valued finite Radon measure $\nu_T = (\nu_{T,1}, \nu_{T,2})$ where each component $\nu_{T,i}$ is a finite signed Radon measure on P. Thus, for every measurable region $E \subseteq P$, the resultant tangential force carried by E is

$$\nu_T(E) = \begin{bmatrix} \nu_{T,1}(E) \\ \nu_{T,2}(E) \end{bmatrix} \in \mathbb{R}^2, \tag{2}$$

2.2 Area measure and pressure fields.

Let dA denote the usual area element on the planar patch P. A *normal pressure field* is a nonnegative function $\rho_N : P \to \mathbb{R}_+$ with units of force per unit area. If $\rho_N \in L^1_+(P)$, which simply means that ρ_N has finite integral over P

$$\rho_N(x) \geq 0 \quad \text{a.e.}, \qquad \int_P \rho_N(x)\, dA < \infty. \tag{3}$$

then it defines a finite normal force measure: the force carried by any region $E \subseteq P$ is given by

$$\nu_N(E) = \int_E \rho_N(x)\, dA. \tag{4}$$

which we also write equivalently as $d\nu_N(x) = \rho_N(x)\, dA$. In that case, ρ_N is called the *density* of the normal force measure ν_N with respect to area[1]. Not every normal force measure admits such a pressure field. For example the Dirac measure $\nu_N = f_N \delta_c$ represents an idealized normal force f_N concentrated at the single point c. It cannot be represented by an ordinary pressure field over area. In this note we adopt the measure-theoretic language (instead of only L^1_+ densities) so that we cover both "classical" pressure fields and idealized concentrated loads.

This notation allows ordinary pressure fields and concentrated loads to be handled by the same integration equations. For a measurable scalar or vector function g, the integral

$$\int_P g(x)\, d\nu_N(x) \tag{5}$$

[1] More generally, the Radon-Nikodym theorem guarantees the existence of a density for measures under suitable assumptions, such as "absolute continuity" which means physically that a subset of zero area (e.g. a point or a line) carries zero normal force. See [1], Theorem 1.30, p. 50

can be interpreted as a force-weighted spatial integral. Whenever ν_N admits a normal pressure field ρ_N,

$$\int_P g(x)\, d\nu_N(x) = \int_P g(x)\rho_N(x)\, dA. \tag{6}$$

For a Dirac measure,

$$\int_P g(x)\, d(f_N\delta_c)(x) = f_N g(c). \tag{7}$$

2.3 Moments of the normal force measure.

The zeroth moment of ν_N is its total measure,

$$m_0(\nu_N) \triangleq \int_P 1\, d\nu_N = \nu_N(P) = f_N. \tag{8}$$

In the present mechanical interpretation, this is the *resultant normal force*. The first moment is

$$m_1(\nu_N) \triangleq \int_P x\, d\nu_N(x) \in \mathbb{R}^2. \tag{9}$$

If $f_N > 0$, the *center of pressure* is the normalized first moment,

$$c[\nu_N] \triangleq \frac{m_1(\nu_N)}{m_0(\nu_N)} \tag{10}$$

$$= \frac{1}{f_N}\int_P x\, d\nu_N(x). \tag{11}$$

Hence knowing the total normal force and the center of pressure is equivalent to knowing the zeroth and first moments of the normal force measure $m_0(\nu_N) = f_N, m_1(\nu_N) = f_N c[\nu_N]$. Unlike the CoP, the pair (m_0, m_1) is linear in ν_N and defined for every measure, including $\nu_N = 0$. We work with these *homogeneous coordinates* throughout, and use the CoP only when $f_N > 0$.

3 Problem formulation

3.1 Planar contact model

Frame. All quantities are coordinates in a fixed frame $\mathcal{R} = (O, \mathcal{B})$ of the contact plane: O is the origin and $\mathcal{B} = (e_1, e_2)$ an orthonormal basis, completed by the contact plane normal e_3. A point of the plane is identified with its coordinate vector $x \in \mathbb{R}^2$. A wrench or a twist is always written in a frame, and its coordinates depend on the reference point (Lemma 3.4); here the reference point is O unless stated otherwise. For $x, y \in \mathbb{R}^2$, the *planar cross product* is the scalar

$$x \times y \triangleq \langle e_3, (x,0) \times (y,0)\rangle = x_1y_2 - x_2y_1 = (Jx)^\top y, \qquad J \triangleq \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \tag{12}$$

Let the planar contact patch $P \subset \mathbb{R}^2$ be compact, and let $\nu_N \in \mathcal{M}_+(P)$ and ν_T be the normal and tangential force measures introduced above. We further assume a known and spatially uniform Coulomb friction coefficient $\mu > 0$. We impose Coulomb friction in measure form by requiring that there exists a measurable density $\gamma_T : P \to \mathbb{R}^2$ of ν_T with respect to ν_N such that

$$d\nu_T(x) = \gamma_T(x)\, d\nu_N(x), \qquad \|\gamma_T(x)\|_2 \le \mu \quad \nu_N\text{-a.e. on } P. \tag{13}$$

We call γ_T the *friction ratio field*. It can be interpreted as the tangential force per unit normal force. The above condition expresses both that tangential friction force can only be carried where

normal force is present and that its local magnitude is bounded by μ[2]. When ν_N admits the normal pressure field ρ_N introduced above, i.e. $d\nu_N = \rho_N\, dA$, the tangential force measure also admits a density with respect to area,

$$\rho_T(x) = \gamma_T(x)\rho_N(x), \tag{14}$$

called the *tangential traction field*. The measure-form Coulomb condition then reduces to the classical local condition

$$\|\rho_T(x)\|_2 \le \mu\rho_N(x) \qquad \text{a.e. on } P. \tag{15}$$

We define the *resultant tangential force* and *resultant torsional moment* about the origin O as

$$f_T \triangleq \nu_T(P) = \int_P \gamma_T(x)\, d\nu_N(x), \qquad \tau_N \triangleq \int_P x \times \gamma_T(x)\, d\nu_N(x). \tag{16}$$

Definition 3.1 (Admissible tangential wrench set)**.** *For a normal force measure $\nu_N \in \mathcal{M}_+(P)$, the admissible tangential wrench set is*

$$\mathcal{X}(\nu_N) \triangleq \left\{ (f_T, \tau_N) \in \mathbb{R}^2 \times \mathbb{R} : \begin{array}{l} \exists\, \gamma_T : P \to \mathbb{R}^2 \text{ measurable such that} \\ \|\gamma_T(x)\|_2 \le \mu \quad \nu_N\text{-a.e.}, \\ f_T = \displaystyle\int_P \gamma_T(x)\, d\nu_N(x), \\ \tau_N = \displaystyle\int_P x \times \gamma_T(x)\, d\nu_N(x) \end{array} \right\}. \tag{17}$$

It is a nonempty, convex, compact subset of $\mathbb{R}^3$, symmetric about 0, containing all resultant tangential wrenches (tangential force and torsional moment about O) that are compatible with the normal force measure ν_N and the local Coulomb friction condition (13).

Proposition 3.2 (Support function of the admissible tangential wrench set)**.** *For any $\nu_N \in \mathcal{M}_+(P)$ and for every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, the support function of $\mathcal{X}(\nu_N)$ is*

$$h_{\mathcal{X}(\nu_N)}(q) = \mu \int_P \|a + bJx\|_2 \; d\nu_N(x). \tag{18}$$

Proof. By definition, $\forall q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, $h_{\mathcal{X}(\nu_N)}(q) = \sup_{(f_T,\tau_N)\in\mathcal{X}(\nu_N)} \left(a^\top f_T + b\tau_N\right)$. For any friction ratio field γ_T,

$$a^\top f_T + b\tau_N = \int_P [a + bJx]^\top \gamma_T(x)\, d\nu_N(x) \le \mu \int_P \|a + bJx\|_2 \; d\nu_N(x), \tag{19}$$

by Cauchy–Schwarz and $\|\gamma_T(x)\|_2 \le \mu$. The bound is attained by the field of magnitude μ aligned with $a + bJx$ where this vector is nonzero, and $\gamma_T^\star(x) = 0$ elsewhere. □

Remark 3.3 (Twists and dissipated power)**.** *The direction $q = (a, b)$ is the coordinate vector of a planar twist about O: a is the velocity of the material point at O and b the angular velocity, so the point at x has velocity $v_q(x) = a + bJx$. The power of the tangential forces under this twist is $\int_P \langle v_q(x), \gamma_T(x)\rangle\, d\nu_N(x) = a^\top f_T + b\tau_N = \langle q, (f_T, \tau_N)\rangle$. Hence $h_{\mathcal{X}(\nu_N)}(q)$ is the largest power that admissible friction can exert under the twist q; since $\mathcal{X}(\nu_N) = -\mathcal{X}(\nu_N)$, it is also the maximal power that friction can dissipate when the contact slides with twist q, which is the value selected by the maximal dissipation principle.*

[2]In particular, $d\nu_T = \gamma_T\, d\nu_N$ implies $\nu_T \ll \nu_N$. Conversely, by the Radon–Nikodym theorem, a tangential measure with $\|\nu_T(E)\|_2 \le \mu\, \nu_N(E)$ for every E has such a density.

Symbol	Meaning	Remark	Lives in
ν_N	normal force measure	true $\nu_N^\star$ is unknown	$\mathcal{M}_+(P)$
(m_0, m_1)	zeroth and first moments of ν_N	$m_1 = f_N c[\nu_N]$ if $f_N > 0$	$\mathbb{R} \times \mathbb{R}^2$
$q = (a, b)$	twist about O (support direction)	prescribed	$\mathbb{R}^2 \times \mathbb{R}$
$\mathcal{M}$	measurement operator	fixed by the sensing model	$\mathcal{M}_+(P) \to \mathbb{R}^r$
y	measurement value	observed data from $(\mathcal{M}, \nu_N)$	$\mathbb{R}^r$
$\Pi_{\mathcal{M}}(y)$	all measures compatible with y	depends on $(\mathcal{M}, y)$	$\subseteq \mathcal{M}_+(P)$
$\underline{h}(q; y, \mathcal{M})$	measurement-conditioned lower bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$
$\overline{h}(q; y, \mathcal{M})$	measurement-conditioned upper bound	depends on $(\mathcal{M}, y, q)$	$\mathbb{R}$
$\mathcal{X}^-(y, \mathcal{M})$, $\mathcal{X}^+(y, \mathcal{M})$	guaranteed and possible wrench sets	Proposition 3.10	$\subseteq \mathbb{R}^3$

Table 1: Notations.

Lemma 3.4 (Change of reference point)**.** *For $o \in \mathbb{R}^2$, let $T_o(f, \tau) \triangleq (f, \tau - o \times f)$ and $S_o(a, b) \triangleq (a + bJo, b)$ (wrench and twist coordinates about o; Varignon's formula). Then T_o and S_o are invertible linear maps and*

(i) $\langle S_o q, T_o w \rangle = \langle q, w \rangle$ for all $q, w \in \mathbb{R}^3$: the power does not depend on the reference point;

(ii) the set of wrenches about o, $\mathcal{X}_o(\nu_N) \triangleq \{(f_T, \int_P (x - o) \times \gamma_T \, d\nu_N)\}$ and $h_{\mathcal{X}_o(\nu_N)}(S_o q) = h_{\mathcal{X}(\nu_N)}(q)$

Proof. (i) $\langle S_o q, T_o w \rangle = (a + bJo)^\top f + b(\tau - o \times f) = a^\top f + b\tau$, since $o \times f = (Jo)^\top f$. (ii) $\int_P (x - o) \times \gamma_T \, d\nu_N = \tau_N - o \times f_T$. Then $h_{\mathcal{X}_o(\nu_N)}(S_o q) = \sup_{w \in \mathcal{X}(\nu_N)} \langle S_o q, T_o w \rangle = h_{\mathcal{X}(\nu_N)}(q)$ by (i), and $S_o^{-1}(a', b) = (a' - bJo, b)$ gives the integral form. □

Lemma 3.5 (Linearity in the normal force)**.** *For $\nu_1, \nu_2 \in \mathcal{M}_+(P)$ and $t \geq 0$, $h_{\mathcal{X}(\nu_1 + \nu_2)} = h_{\mathcal{X}(\nu_1)} + h_{\mathcal{X}(\nu_2)}$ and $h_{\mathcal{X}(t\nu_1)} = t \, h_{\mathcal{X}(\nu_1)}$. Consequently $\mathcal{X}(\nu_1 + \nu_2) = \mathcal{X}(\nu_1) \oplus \mathcal{X}(\nu_2)$ (Minkowski sum), $\mathcal{X}(t\nu_1) = t \, \mathcal{X}(\nu_1)$ and $\mathcal{X}(0) = \{0\}$.*

Proof. The first claims follow from Proposition 3.2. The support function of a Minkowski sum is the sum of the support functions, and nonempty compact convex sets with the same support function are equal [4, Section 13]. □

3.2 Measurement-conditioned support bounds

We focus here on the case where we do not know the *true* force measure $\nu_N^\star$ but we have only access to some measurements of it (e.g. resultant normal force, CoP, etc.). The goal is to investigate what can be known about the support of $\mathcal{X}(\nu_N^\star)$ given partial measurements of $\nu_N^\star$.

Definition 3.6 (Measurement operator)**.** *A measurement operator is a map $\mathcal{M} : \mathcal{M}_+(P) \to \mathbb{R}^r$ of the form*

$$\mathcal{M}(\nu_N) = \begin{bmatrix} \int_P \phi_1(x) \, d\nu_N(x) \\ \vdots \\ \int_P \phi_r(x) \, d\nu_N(x) \end{bmatrix}, \qquad \forall \nu_N \in \mathcal{M}_+(P), \tag{20}$$

where the prescribed bounded Borel functions $\phi_j : P \to \mathbb{R}$ are called the measurement functions. For example, $\phi = 1$ measures the resultant normal force f_N, $\phi(x) = x_1$ and $\phi(x) = x_2$ its first moment $m_1 = f_N c$ (CoP in homogeneous coordinates) In the following, we denote a measurement value (or observation) induced by the (unknown) true force measure $\nu_N^\star$ as $y = \mathcal{M}(\nu_N^\star) \in \mathbb{R}^r$.

Definition 3.7 (Measurement-compatible normal force family)**.** *Given a measurement operator $\mathcal{M} : \mathcal{M}_+(P) \to \mathbb{R}^r$ and the observation $y \in \mathbb{R}^r$, the set of normal force measures compatible with y is the preimage*

$$\Pi_{\mathcal{M}}(y) \triangleq \mathcal{M}^{-1}[\{y\}] = \{\nu_N \in \mathcal{M}_+(P) : \mathcal{M}(\nu_N) = y\} \tag{21}$$

It contains all the normal force measures that are indistinguishable under the measurement y

Given an observation y of the unknown true force measure $\nu_N^\star$, we can compute measurement-conditioned lower/upper bounds on the support function $h_{\mathcal{X}(\nu_N^\star)}$ by optimizing over $\Pi_{\mathcal{M}}(y)$.

Definition 3.8 (Measurement-conditioned support bounds)**.** *Let $\mathcal{M}$ be a measurement operator and $y \in \mathbb{R}^r$ an observation. For a twist $q \in \mathbb{R}^2 \times \mathbb{R}$, the measurement-conditioned bounds in direction q are*

$$\underline{h}(q; y, \mathcal{M}) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{X}(\nu_N)}(q) = \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJx\|_2 \, d\nu_N(x), \tag{22}$$

$$\overline{h}(q; y, \mathcal{M}) \triangleq \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} h_{\mathcal{X}(\nu_N)}(q) = \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mu \int_P \|a + bJx\|_2 \, d\nu_N(x), \tag{23}$$

and the guaranteed and possible wrench sets are

$$\mathcal{X}^-(y, \mathcal{M}) \triangleq \bigcap_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{X}(\nu_N), \qquad \mathcal{X}^+(y, \mathcal{M}) \triangleq \bigcup_{\nu_N \in \Pi_{\mathcal{M}}(y)} \mathcal{X}(\nu_N). \tag{24}$$

The objective of both problems in (22) is linear in ν_N, and the constraints $\int_P \phi_j \, d\nu_N = y_j$ are linear moment constraints: *the bounds are Generalized Moment Problems (GMP)* [2], for every measurement operator.

Assumption 3.9 (Mass control)**.** *There exists $\lambda \in \mathbb{R}^r$ such that $\sum_j \lambda_j \phi_j(x) \geq 1$ for every $x \in P$.*

Under Assumption 3.9, every $\nu_N \in \Pi_{\mathcal{M}}(y)$ satisfies $\nu_N(P) \leq \lambda^\top y$, so the compatible family has bounded mass. This assumption keeps the upper bounds finite: without it, force could be hidden where all the measurement functions vanish. For example, consider $P = [0, L] \times [0, \ell]$, $r = 1$, $\phi(x) = x_1$ (so $\mathcal{M}(\nu_N) = \int_P x_1 \, d\nu_N$), and the observation $y = 0$. For every $t \geq 0$ and $x_2 \in [0, \ell]$, the measure $\nu_t = t \, \delta_{(0,x_2)}$ is compatible. Then take $q = (a, 0)$ with $a \neq 0$; the integrand is $\|a + bJx\|_2 = \|a\|_2$ everywhere, so $h_{\mathcal{X}(\nu_t)}(q) = \mu\|a\|_2 \, t$. Letting $t \to \infty$ gives $\overline{h}(q; 0, \mathcal{M}) = +\infty$. To prevent this pathology, all measurement operators studied below measure the resultant normal force (directly or as sum of the taxel forces) and satisfy Assumption 3.9.

Proposition 3.10 (Bounds and wrench sets)**.** *Let $\mathcal{M}$ (with Assumption 3.9) and $y \in \mathbb{R}^r$.*

(i) The possible set $\mathcal{X}^+(y, \mathcal{M})$ is convex, and its support function is $\overline{h}(\cdot; y, \mathcal{M})$.

(ii) The guaranteed set is $\mathcal{X}^-(y, \mathcal{M}) = \{w : \langle q, w \rangle \leq \underline{h}(q; y, \mathcal{M}) \ \forall q\}$. Its support function is the closed convex envelope of $\underline{h}(\cdot; y, \mathcal{M})$, which is in general strictly below $\underline{h}$.

(iii) $y \mapsto \overline{h}(q; y, \mathcal{M})$ is concave and $y \mapsto \underline{h}(q; y, \mathcal{M})$ is convex on the set of observations $\mathcal{M}(\mathcal{M}_+(P))$, and both are positively homogeneous.

Proof. (i) Mixing two compatible measures gives a compatible measure ($\Pi_{\mathcal{M}}(y)$ is convex), and $\mathcal{X}$ is linear in the measure (Lemma 3.5). So a mixture of two wrenches of $\mathcal{X}^+$ is a wrench of the mixed measure, hence belongs to $\mathcal{X}^+$. The support function of a union is the supremum of the support functions, which is $\overline{h}$. (ii) A wrench belongs to $\mathcal{X}(\nu_N)$ for every compatible ν_N if and only if $\langle q, w \rangle \leq h_{\mathcal{X}(\nu_N)}(q)$ for all q and all compatible ν_N, i.e., $\langle q, w \rangle \leq \underline{h}(q; y, \mathcal{M})$ for all q (a closed convex set is the intersection of its supporting half-spaces). The support function of the

set defined by these inequalities is the closed convex envelope of $\underline{h}$ [4, Section 13]. (iii) Mixing measures compatible with y_1 and y_2 gives a measure compatible with the mixed observation, and $h_{\mathcal{X}(\nu_N)}(q)$ is linear in ν_N. Taking the supremum (resp. infimum) over the two measures gives concavity (resp. convexity). Positive homogeneity follows by scaling the measure. □

Remark 3.11 (How to use the bounds)**.** *A candidate tangential wrench w (about O) is carried by every compatible normal force measure, i.e., $w \in \mathcal{X}^-(y, \mathcal{M})$, if and only if $\langle q, w\rangle \le \underline{h}(q; y, \mathcal{M})$ for all q. Conversely, if $\langle q, w\rangle > \overline{h}(q; y, \mathcal{M})$ for some q, then $w \notin \mathcal{X}^+(y, \mathcal{M})$: no compatible measure carries w, and the contact slips whatever the actual pressure distribution. By Proposition 3.10(i), the second test is a convex membership test. The sets are closed: a wrench with $\langle q, w\rangle = h_{\mathcal{X}(\nu_N)}(q)$ for some q lies on the boundary of $\mathcal{X}(\nu_N)$, where the contact is at the onset of sliding with twist q (maximal dissipation). A strict inequality, or a margin, certifies the interior. Section 9 turns both tests into convex programs along a load direction*[3]*.*

3.3 General structure of the bounds

For $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$, we write

$$g(x; q) \triangleq \|a + bJx\|_2 \,, \tag{25}$$

the speed of the material point at x under the twist q, so that, by Proposition 3.2, $h_{\mathcal{X}(\nu_N)}(q) = \mu \int_P g(x; q)\, d\nu_N(x)$. For $b \neq 0$, we can write $a + bJx = bJ(x - p)$ where $p \triangleq Ja/b$ is the instantaneous center of rotation (ICR), so that $g(x; q) = |b|\, \|x - p\|_2$ [4]. For $b = 0$, $g(x; q) = \|a\|_2$.

Lemma 3.12 (Properties of the integrand)**.** *For every $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$:*

(i) $x \mapsto g(x; q)$ is convex on $\mathbb{R}^2$;

(ii) for all $x, x' \in \mathbb{R}^2$, $|g(x; q) - g(x'; q)| \le |b|\, \|x - x'\|_2$;

(iii) for every $\nu_N \in \mathcal{M}_+(P)$, we have $h_{\mathcal{X}(\nu_N)}(q) \ge \mu\, \|m_0(\nu_N)\, a + bJm_1(\nu_N)\|_2$. For $f_N > 0$, it reads $h_{\mathcal{X}(\nu_N)}(q) \ge \mu f_N\, g(c[\nu_N]; q)$: the maximal dissipated power is at least that of the whole normal force concentrated at the CoP.

Proof. (i) $g(\cdot; q)$ is a norm composed with an affine map. (ii) By the reverse triangle inequality and $\|J\|_2 = 1$. (iii) By the triangle inequality for vector-valued integrals, $\int_P \|a + bJx\|_2\, d\nu_N \ge \|\int_P (a + bJx)\, d\nu_N\|_2 = \|m_0 a + bJm_1\|_2$. □

The bounds of two measurement operators can only be compared if their observations come from the same normal force measure. This leads to an order on the information carried by measurement operators, which we can formalize (Yann).

Definition 3.13 (Information preorder)**.** *A measurement operator $\mathcal{M}$ is at least as informative as $\mathcal{M}'$, written $\mathcal{M} \succeq \mathcal{M}'$, if*

$$\Pi_{\mathcal{M}}(\mathcal{M}(\nu_N)) \subseteq \Pi_{\mathcal{M}'}(\mathcal{M}'(\nu_N)) \qquad \text{for every } \nu_N \in \mathcal{M}_+(P), \tag{26}$$

i.e., every measure that $\mathcal{M}$ cannot distinguish from ν_N is also indistinguishable from ν_N under $\mathcal{M}'$. The concatenation $\mathcal{M} \vee \mathcal{M}' \triangleq (\mathcal{M}, \mathcal{M}')$ is the operator with measurement functions $(\phi_1, \ldots, \phi_r, \phi'_1, \ldots, \phi'_{r'})$.

[3] Because all support functions are positively homogeneous in q, it is sufficient to consider one normalized representative of each nonzero ray.

[4] Note that this is the classical planar sliding form of the dissipation: the sliding speed of each contact point is its distance to the ICR times the angular rate [5, Eq. 7], and the limit surface is the maximum-power envelope of this quantity [6, §8].

Measurement	Observed data	Lower bound	Upper bound
$\mathcal{M}_0$ (Sec. 4)	$y_0 = (f_N, m_1)$	closed form	LP over the vertices of P (closed form via triangles)
$\mathcal{M}_F$ (Sec. 6)	$y_F = (F_1, \dots, F_r)$	closed form: $\sum_i$ of minima over the taxels	closed form: $\sum_i$ of maxima over the taxel corners
$\mathcal{M}_T$ (Sec. 7)	$y_T = (y_0, y_F)$	SOCP	LP over the taxel corners

Table 2: Measurement operators and structure of the resulting bounds. All bounds are GMPs; the CoP constraint of $\mathcal{M}_T$ couples the taxels.

Proposition 3.14 (Properties of the information preorder)**.** *(i) $\succeq$ is a preorder (reflexive and transitive). It is not antisymmetric, and not total.*

(ii) $\mathcal{M} \succeq \mathcal{M}'$ if and only if there is a matrix $A \in \mathbb{R}^{r' \times r}$ with $\phi'_j(x) = \sum_k A_{jk}\phi_k(x)$ for every $x \in P$ and every j, i.e., $\mathcal{M}' = A\,\mathcal{M}$.

(iii) $\mathcal{M} \vee \mathcal{M}' \succeq \mathcal{M}$ and $\mathcal{M} \vee \mathcal{M}' \succeq \mathcal{M}'$, with $\Pi_{\mathcal{M}\vee\mathcal{M}'}((y, y')) = \Pi_{\mathcal{M}}(y) \cap \Pi_{\mathcal{M}'}(y')$.

Proof. (i) Reflexivity and transitivity are inclusions of compatible families. The preorder is not antisymmetric: scaling all measurement functions by 2 gives a different operator with the same compatible families. It is not total: $\mathcal{M}_0$ and $\mathcal{M}_F$ of the next sections are incomparable (see (ii)). (ii) ($\Leftarrow$) If $\mathcal{M}' = A\mathcal{M}$, a measure with the same $\mathcal{M}$-value as ν_N has the same $\mathcal{M}'$-value. ($\Rightarrow$) Write $\Phi = (\phi_k)_k$ and $\Phi' = (\phi'_j)_j$. A matrix A with $A\Phi = \Phi'$ on P exists as soon as every linear relation between values of Φ also holds for Φ', i.e., $\sum_i \alpha_i \Phi(x_i) = 0 \Rightarrow \sum_i \alpha_i \Phi'(x_i) = 0$ for all finite families of points. Such a relation says that the two measures $\sigma^{\pm} = \sum_{i:\pm\alpha_i>0} |\alpha_i| \delta_{x_i}$ have the same $\mathcal{M}$-value. Then σ^- is compatible with $\mathcal{M}(\sigma^+)$, hence, since $\mathcal{M} \succeq \mathcal{M}'$, with $\mathcal{M}'(\sigma^+)$: $\mathcal{M}'(\sigma^+) = \mathcal{M}'(\sigma^-)$, which is the relation for Φ'. (iii) Immediate from the definitions. □

The characterization (ii) is pointwise: functions that agree almost everywhere with respect to area are not equivalent, since point loads see their values at single points. The number r of measurement functions is not monotone with respect to $\succeq$, because measurement functions can be redundant. The operator without measurement functions ($r = 0$, $\Pi = \mathcal{M}_+(P)$) is below every operator; its bounds are 0 and $+\infty$, and it violates Assumption 3.9.

Theorem 3.15 (Monotonicity with respect to information)**.** *Let $\mathcal{M} \succeq \mathcal{M}'$, $\nu_N \in \mathcal{M}_+(P)$, $y = \mathcal{M}(\nu_N)$ and $y' = \mathcal{M}'(\nu_N)$. Then, for every $q \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\underline{h}(q; y', \mathcal{M}') \le \underline{h}(q; y, \mathcal{M}) \le h_{\mathcal{X}(\nu_N)}(q) \le \overline{h}(q; y, \mathcal{M}) \le \overline{h}(q; y', \mathcal{M}'), \tag{27}$$

and $\mathcal{X}^-(y', \mathcal{M}') \subseteq \mathcal{X}^-(y, \mathcal{M}) \subseteq \mathcal{X}(\nu_N) \subseteq \mathcal{X}^+(y, \mathcal{M}) \subseteq \mathcal{X}^+(y', \mathcal{M}')$. This applies in particular to the true measure $\nu_N^\star$.

Proof. $\nu_N \in \Pi_{\mathcal{M}}(y) \subseteq \Pi_{\mathcal{M}'}(y')$. An infimum (resp. supremum, intersection, union) over a smaller family is larger (resp. smaller, larger, smaller). □

Table 2 summarizes the measurement operators studied in the remainder of the report and the structure of the resulting problems, and Figure 1 illustrates what each of them observes.

4 $\mathcal{M}_0$ case (coarse measurement)

We now assume that the available measurements consist only of the resultant normal force and first moment. **Known:** P, μ, f_N and m_1. **Unknown:** how $\nu_N^\star$ is distributed over P. In this section P is convex.

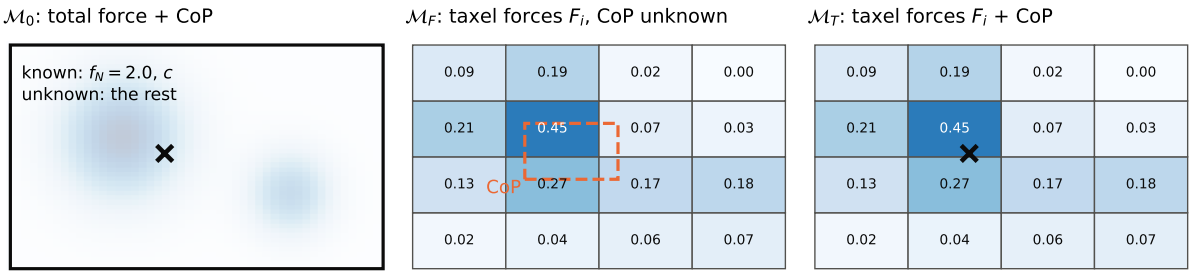


Figure 1: What each measurement operator observes for the same contact (4×4 taxels). $\mathcal{M}_0$: only f_N and the CoP, the pressure distribution (faded) is unknown. $\mathcal{M}_F$: the taxel forces F_i (cells filled with the mean pressure $F_i/|P_i|$); the CoP is only known to lie in the dashed rectangle $\bar{c}_F + R_0$, where $\bar{c}_F = \frac{1}{f_N}\sum_i F_i m_i$ is the force-weighted average of the taxel centers and $R_0 = [-\frac{w_x}{2}, \frac{w_x}{2}] \times [-\frac{w_y}{2}, \frac{w_y}{2}]$. Indeed, the local first moments lie in $F_i\overline{P_i}$, so the CoP lies in $\sum_i \frac{F_i}{f_N}\overline{P_i} = \bar{c}_F + R_0$, and point loads in the taxels reach every point of it. It has the diameter D of one taxel, whatever the forces, and an observed CoP outside it would contradict the taxel forces; the bounds do not use this set. $\mathcal{M}_T$: taxel forces and CoP.

Definition 4.1 (Coarse normal force measurement)**.** *Given the true (unknown) normal force measure $\nu_N^\star \in \mathcal{M}_+(P)$, the coarse measurement is its zeroth and first moments*

$$y_0 = \mathcal{M}_0(\nu_N^\star) \triangleq \left(\int_P d\nu_N^\star(x),\ \int_P x\, d\nu_N^\star(x) \right) = (f_N, m_1). \tag{28}$$

Its measurement functions are $\phi_1(x) = 1$, $\phi_2(x) = x_1$, $\phi_3(x) = x_2$, and it satisfies Assumption 3.9 with $\lambda = (1, 0, 0)$.

Theorem 4.2 (Exact coarse lower bound)**.** *Given the coarse observation $y_0 = (f_N, m_1)$, for any $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,*

$$\boxed{\underline{h}(q; y_0, \mathcal{M}_0) = \mu \, \|f_N a + bJm_1\|_2} \tag{29}$$

For $f_N > 0$ this is $\mu f_N \, g(c; q)$: maximal dissipated power of normal force concentrated at the CoP.

Proof. By Lemma 3.12(iii), $h_{\mathcal{X}(\nu_N)}(q) \geq \mu\|f_N a + bJm_1\|_2$ for every $\nu_N \in \Pi_{\mathcal{M}_0}(y_0)$. If $f_N = 0$, then $\Pi_{\mathcal{M}_0}(y_0) = \{0\}$ and both sides vanish. Otherwise $c \in P$, so $f_N\delta_c \in \Pi_{\mathcal{M}_0}(y_0)$, and it attains the value. □

Corollary 4.3 (Exact guaranteed set; no torsion about the CoP)**.** *Let $f_N > 0$ and $c = m_1/f_N$. Then*

$$\mathcal{X}^-(y_0, \mathcal{M}_0) = \mathcal{X}(f_N\delta_c) = \{(f, c \times f) :\ \|f\|_2 \leq \mu f_N\}. \tag{30}$$

In particular, in CoP coordinates (Lemma 3.4), $T_c\mathcal{X}^-(y_0, \mathcal{M}_0) = f_N\, B(0, \mu) \times \{0\}$: coarse measurement guarantees the friction for the tangential force, but no torsional moment about the CoP.

Proof. We show both inclusions. ($\subseteq$) The point load $f_N\delta_c$ is compatible, so $\mathcal{X}^- \subseteq \mathcal{X}(f_N\delta_c)$. With all the friction at c, its wrenches are $(f, c \times f)$ with $\|f\|_2 \leq \mu f_N$. ($\supseteq$) Let $\|f\|_2 \leq \mu f_N$ and let ν_N be any compatible measure. Mobilizing the same friction ratio f/f_N, of norm at most μ, at every contact point is admissible. Its resultant force is f, and its moment is $\int x\, d\nu_N \times f/f_N = m_1 \times f/f_N = c \times f$. So $(f, c \times f) \in \mathcal{X}(\nu_N)$ (compare Remark 4.2 of Yann's draft). □

Theorem 4.4 (Coarse upper bound). *The supremum $\overline{h}(q; y_0, \mathcal{M}_0)$ is attained by a measure $\nu_N^+ \in \Pi_{\mathcal{M}_0}(y_0)$ supported on at most three points, i.e.,*

$$\nu_N^+ = \sum_{j=1}^{s} \lambda_j \delta_{x_j}, \qquad s \le 3, \qquad x_j \in P, \quad \lambda_j \ge 0, \qquad \sum_{j=1}^{s} \lambda_j = f_N, \qquad \sum_{j=1}^{s} \lambda_j x_j = m_1. \tag{31}$$

Proof. This is Proposition 4.4 of Yann's draft with $C = B(0, \mu)$ (Remark 4.6). It is an instance of the classical fact that extreme points of a set of measures defined by three linear moment constraints have at most three atoms [3, 2]. A direct argument has three steps (for $f_N = 0$ there is nothing to prove). *1. Geometric reformulation.* Let $\pi = \nu_N / f_N$ and $\Phi(x) = (x, g(x; q)) \in \mathbb{R}^3$. The constraints say that the first two components of $\int \Phi \, d\pi$ equal the CoP c, and the third is the objective divided by μf_N. Such averages fill the compact convex set $K = \operatorname{conv} \Phi(P)$, so the bound is $\mu f_N t^\star$, where $(c, t^\star)$ is the highest point of K above c. *2. The maximizer lies on a face.* The point $(c, t^\star)$ is on the boundary of K, so it lies on a supporting plane H, and $K \cap H = \operatorname{conv}(\Phi(P) \cap H)$. *3. Carathéodory.* $K \cap H$ has dimension at most two, so $(c, t^\star)$ is a convex combination $\sum_j \beta_j \Phi(x_j)$ of at most three points of $\Phi(P)$. The measure $f_N \sum_j \beta_j \delta_{x_j}$ is compatible and attains the bound. □

Corollary 4.5 (Polygonal contact patch). *Let P be a convex polygon with vertices $v_1, \dots, v_K$. Then*

$$\boxed{\overline{h}(q; y_0, \mathcal{M}_0) = \mu \max_{\lambda \in \mathbb{R}_+^K} \left\{ \sum_{k=1}^{K} \lambda_k \| a + bJ v_k \|_2 : \sum_{k=1}^{K} \lambda_k = f_N, \ \sum_{k=1}^{K} \lambda_k v_k = m_1 \right\}} \tag{32}$$

This is a linear program with three equality constraints, so it has an optimal solution with at most three nonzero weights.

Proof. Moving mass to the vertices. By Theorem 4.4, the supremum is attained by a measure with at most three atoms, so it suffices to consider such measures. Split the mass λ at an atom $x = \sum_k \alpha_k v_k$ (a convex combination of the vertices) into the masses $\lambda \alpha_k$ at the vertices v_k. This keeps f_N and m_1, and by convexity of $g(\cdot; q)$ (Lemma 3.12(i)) it does not decrease the objective. So the supremum is at most the value of the linear program. *Conversely,* a feasible λ gives the compatible measure $\sum_k \lambda_k \delta_{v_k}$, whose objective is the value of the linear program. *Three weights.* A linear program with three equality constraints has a basic optimal solution with at most three nonzero entries. □

Remark 4.6 (Connection with Yann's formulation). *Theorem 4.4 and Corollary 4.5 are the Coulomb case $C = B(0, \mu)$ of Proposition 4.4 in Yann's draft, with the same reference point. With his notation $x^\perp = -Jx$, his moment $m_N = -\int \langle \rho_T, x^\perp \rangle$ is our τ_N, his twist (ω_N, v_T) is our $q = (v_T, \omega_N)$, and his integrand is $s_C(\omega_N x^\perp - v_T) = \mu \| v_T + \omega_N J x \|_2 = \mu \, g(x; q)$. His set $f_N \, D(c^\perp)$ is $\mathcal{X}^+(y_0, \mathcal{M}_0)$ (components reordered), so that $f_N \, s_{D(c^\perp)}(-[\omega_N, v_T]) = \overline{h}(q; y_0, \mathcal{M}_0)$, and the concavity of $a \mapsto s_{D(a)}$ is Proposition 3.10(iii). His special cases are the CoP at a vertex of P (a single point load, both bounds equal $\mu f_N g(c; q)$) and $\omega_N = 0$ (both bounds equal $\mu f_N \| v_T \|_2$). As he notes, $D(\cdot)$ is the place to add priors on the distribution; the taxel measurements of the next sections are such priors.*

Example 4.7 (Segment patch). *Let $P = [0, L] \times \{0\}$, $c = \left(\frac{L}{2}, 0\right)$, $\nu_N^{(1)} = f_N \delta_c$ and $\nu_N^{(2)} = \frac{f_N}{2} \delta_{(0,0)} + \frac{f_N}{2} \delta_{(L,0)}$. Both have $\mathcal{M}_0(\nu_N^{(1)}) = \mathcal{M}_0(\nu_N^{(2)}) = \left(f_N, f_N \left(\frac{L}{2}, 0\right)\right)$. For the unit rotation about the CoP, $q_c = (-Jc, 1)$, so that $g(x; q_c) = \| x - c \|_2$,*

$$h_{\mathcal{X}(\nu_N^{(1)})}(q_c) = 0 \quad \neq \quad h_{\mathcal{X}(\nu_N^{(2)})}(q_c) = \mu f_N \frac{L}{2} = \overline{h}(q_c; y_0, \mathcal{M}_0). \tag{33}$$

The results of this section are recovered in Section 7 as the special case of a single taxel $P_1 = P$.

5 Taxel grids: setting and common lemmas

We now assume that a tactile array measures the normal force carried by each taxel, with or without the coarse measurement. Tactile arrays are grids of rectangular taxels, and we restrict to them. Other cell shapes need barycentric weights in Lemma 5.5 instead of bilinear ones, and are not detailed. This section fixes the setting and collects the lemmas used in Sections 6 and 7.

Definition 5.1 (Taxel grid)**.** *Let $P = [0, L] \times [0, \ell]$ be the sensor area. For $k_x, k_y \geq 1$, the grid $G(k_x, k_y)$ divides P into $r = k_x k_y$ taxels $P_1, \dots, P_r$: the rectangles $[\mathrm{lo}_i, \mathrm{hi}_i)$ (componentwise half-open) of size $w_x \times w_y = \frac{L}{k_x} \times \frac{\ell}{k_y}$, those of the last column and row being closed on their outer side, so that every point of P belongs to exactly one taxel. We write $\overline{P_i} = [\mathrm{lo}_i, \mathrm{hi}_i]$ for the closed taxel, m_i for its center, $v_{i1}, \dots, v_{i4}$ for its corners, and $D = \sqrt{w_x^2 + w_y^2}$ for the taxel diameter.*

Definition 5.2 (Taxel force measurement)**.** *Given the true unknown normal force measure $\nu_N^\star$ and a taxel grid, the taxel force measurement is*

$$y_F = \mathcal{M}_F(\nu_N^\star) \triangleq (\nu_N^\star(P_1), \dots, \nu_N^\star(P_r)), \tag{34}$$

with measurement functions $\phi_i = \mathbf{1}_{P_i}$, $i = 1, \dots, r$. The measured taxel forces are denoted $F_i \triangleq \nu_N^\star(P_i)$, and $f_N = \sum_i F_i$. Since $\sum_i \mathbf{1}_{P_i} = 1$, $\mathcal{M}_F$ satisfies Assumption 3.9. For the grid $G(k_x, k_y)$ we write $\mathcal{M}_F(k_x, k_y)$ (e.g. Yann's suggested grid is $\mathcal{M}_F(2^{p_x}, 2^{p_y})$).

Definition 5.3 (Combined coarse and taxel force measurement)**.** *The combined measurement is the concatenation $\mathcal{M}_T \triangleq \mathcal{M}_0 \vee \mathcal{M}_F$,*

$$y_T = \mathcal{M}_T(\nu_N^\star) = (f_N, m_1, F_1, \dots, F_r), \qquad \Pi_{\mathcal{M}_T}(y_T) = \Pi_{\mathcal{M}_0}(y_0) \cap \Pi_{\mathcal{M}_F}(y_F). \tag{35}$$

We write $\mathcal{M}_T(k_x, k_y) \triangleq \mathcal{M}_0 \vee \mathcal{M}_F(k_x, k_y)$

Proposition 5.4 (Information chains)**.** *(i) $\mathcal{M}_F(k_x, k_y) \succeq \mathcal{M}_F(k'_x, k'_y)$ if and only if k'_x divides k_x and k'_y divides k_y. The same holds for $\mathcal{M}_T$. In particular, dyadic grids form chains, while 3×3 and 4×4 grids are not comparable. Moreover $\mathcal{M}_F(1, 1)$ measures the resultant force f_N only, and $\mathcal{M}_T(1, 1)$ and $\mathcal{M}_0$ are equivalent.*

(ii) $\mathcal{M}_T \succeq \mathcal{M}_0$ and $\mathcal{M}_T \succeq \mathcal{M}_F$. If $r \geq 2$, $\mathcal{M}_0$ and $\mathcal{M}_F$ are not comparable.

Proof. Recall that $\mathcal{M} \succeq \mathcal{M}'$ if and only if every measurement function of $\mathcal{M}'$ is a linear combination of those of $\mathcal{M}$ (Proposition 3.14). (i) If $k'_x \mid k_x$ and $k'_y \mid k_y$, each coarse taxel is a union of fine taxels, so its indicator is the sum of the indicators of the fine taxels it contains. Conversely, suppose for example that $k'_x \nmid k_x$. Then some edge between two coarse taxels lies strictly inside a fine taxel. The indicator of a coarse taxel jumps across this edge, whereas the fine indicators are constant there and $1, x_1, x_2$ are continuous, so it is not a linear combination of them. The case $k'_y \nmid k_y$ is the same. (ii) $\mathcal{M}_T$ has the measurement functions of both. If $r \geq 2$, a taxel indicator is not affine, so $\mathcal{M}_0 \not\succeq \mathcal{M}_F$, and x_1 is not constant on a taxel, so $\mathcal{M}_F \not\succeq \mathcal{M}_0$. □

Lemma 5.5 (Concentration and spreading)**.** *Let $\nu_N \in \mathcal{M}_+(P)$ have taxel forces $F_i = \nu_N(P_i)$ and local first moments $u_i \triangleq \int_{P_i} x \, d\nu_N(x)$, so that $\sum_i F_i = f_N$ and $\sum_i u_i = m_1$. There are weights $\lambda_{ik} \geq 0$ with $\sum_k \lambda_{ik} = F_i$ and $\sum_k \lambda_{ik} v_{ik} = u_i$ (and $\lambda_{ik} = 0$ if $F_i = 0$) such that, for every twist $q = (a, b)$,*

$$\underbrace{\mu \sum_{i : F_i > 0} F_i \, g\Big(\tfrac{u_i}{F_i}; q\Big)}_{h_{\mathcal{X}(\nu_N^c)}(q)} \;\leq\; h_{\mathcal{X}(\nu_N)}(q) \;\leq\; \underbrace{\mu \sum_{i,k} \lambda_{ik} \, g(v_{ik}; q)}_{h_{\mathcal{X}(\nu_N^s)}(q)}. \tag{36}$$

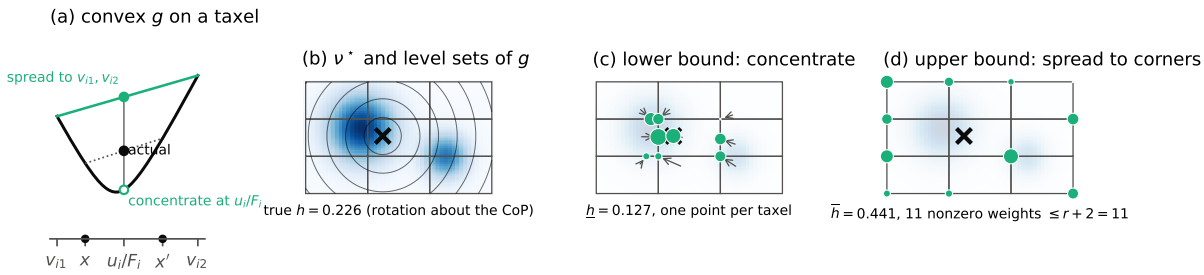


Figure 2: Concentrate vs. spread (Lemma 5.5, Eq. (36)), 3×3 taxels, unit rotation about the CoP $q_c = (-Jc, 1)$, so $g(x; q_c) = \|x - c\|_2$. (a) For two masses at x, x' of a taxel, with local CoP u_i/F_i: concentrating at u_i/F_i lowers the objective (Jensen), spreading to the corners v_{i1}, v_{i2} raises it (chord). (b) True measure and level sets of g. (c) Minimizer of Theorem 7.1: one Dirac per taxel, pulled from the local CoPs of the taxels toward the CoP (arrows). (d) Maximizer of Theorem 7.2: weights on the taxel corners, pushed away from the CoP, with at most $r+2$ nonzero weights. Marker areas are proportional to force.

The outer terms are the values of $h_{\mathcal{X}(\cdot)}(q)$ *for the* concentrated *load* $\nu_N^c = \sum_i F_i \delta_{u_i/F_i}$ *and the* spread *load* $\nu_N^s = \sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$. *Both put the force* F_i *in the closed taxel* $\overline{P_i}$ *and have the first moment* m_1 *of* ν_N. *The same holds about any reference point* o *for* $h_{\mathcal{X}_o(\nu_N)}$, *with* $g(\cdot - o; q)$ *in place of* $g(\cdot; q)$.

Proof. By Proposition 3.2,

$$h_{\mathcal{X}(\nu_N)}(q) = \mu \sum_i \int_{P_i} g(x; q)\, d\nu_N(x),$$

and $g(\cdot; q)$ is convex (Lemma 3.12(i)). Fix a taxel with $F_i > 0$. *Concentration.* $\nu_N|_{P_i}/F_i$ is a probability measure with mean u_i/F_i, so Jensen's inequality gives $\int_{P_i} g\, d\nu_N \geq F_i\, g(u_i/F_i; q)$. *Spreading.* For $x \in \overline{P_i}$, let $\xi = (x_1 - \mathrm{lo}_{i,1})/w_x$ and $\eta = (x_2 - \mathrm{lo}_{i,2})/w_y$, and let $\alpha_{ik}(x)$ be the bilinear weights of the four corners, $(1-\xi)(1-\eta)$, $\xi(1-\eta)$, $\xi\eta$, $(1-\xi)\eta$ (counterclockwise from lo_i). They are nonnegative, sum to one, and satisfy $\sum_k \alpha_{ik}(x) v_{ik} = x$. By convexity, $g(x; q) \leq \sum_k \alpha_{ik}(x)\, g(v_{ik}; q)$. Integrating against $\nu_N|_{P_i}$ gives the right inequality with $\lambda_{ik} = \int_{P_i} \alpha_{ik}\, d\nu_N$. Integrating $\sum_k \alpha_{ik} = 1$ and $\sum_k \alpha_{ik} v_{ik} = x$ in the same way gives $\sum_k \lambda_{ik} = F_i$ and $\sum_k \lambda_{ik} v_{ik} = u_i$. Summing over the taxels proves the claim. About o, use $h_{\mathcal{X}_o(\nu_N)}(a', b) = \mu \int g(x - o; (a', b))\, d\nu_N$ (Lemma 3.4) and translate x, u_i/F_i and v_{ik} by $-o$. □

Interpretation. The lemma replaces the unknown measure, taxel by taxel, by finitely many point loads that keep the data of the measurement (F_i and m_1), and brackets its friction power in every direction (Figure 2).

- *Concentrate* (left inequality): placing the force of each taxel at its local CoP u_i/F_i can only decrease the friction power. This gives the lower bounds of $\mathcal{M}_F$ (Theorem 6.1) and $\mathcal{M}_T$ (Theorem 7.1); for $\mathcal{M}_0$ it is the single point load at the CoP (Theorem 4.2).
- *Spread* (right inequality): splitting the force of each taxel among its corners, with the weights of the lemma, can only increase it. This gives the upper bounds of $\mathcal{M}_F$ and $\mathcal{M}_T$ (Theorems 6.1 and 7.2); for $\mathcal{M}_0$ it is the reduction to the vertices of P (Corollary 4.5), i.e., Yann's reduction to the extreme points (his Eq. (33) and Proposition 4.4), applied taxel by taxel.

6 $\mathcal{M}_F$ case (taxel forces)

Known: the taxel forces F_i. **Unknown:** the distribution of $\nu_N^\star$ inside each taxel, hence also its CoP. Without the CoP constraint, the taxels decouple and both bounds have closed forms.

Theorem 6.1 (Exact bounds for taxel force measurements)**.** *On a taxel grid, for every* $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$,

$$\boxed{\underline{h}(q; y_F, \mathcal{M}_F) = \mu \sum_{i=1}^{r} F_i \inf_{x \in P_i} \|a + bJx\|_2, \qquad \overline{h}(q; y_F, \mathcal{M}_F) = \mu \sum_{i=1}^{r} F_i \sup_{x \in P_i} \|a + bJx\|_2.} \tag{37}$$

The supremum equals $\max_{k=1,\dots,4} \|a + bJv_{ik}\|_2$*, the maximum over the corners of taxel* i*. For* $b \neq 0$*, with the instantaneous center of rotation* $p = Ja/b$*, the bounds are* $\mu|b| \sum_i F_i\, d_i^{\min}$ *and* $\mu|b| \sum_i F_i\, d_i^{\max}$*, where* $d_i^{\min}$ *and* $d_i^{\max}$ *are the smallest and largest distances from* p *to the taxel. With* $\Delta_i = m_i - p$*,*

$$d_i^{\min} = \left\| \left(\left(|\Delta_{i,1}| - \tfrac{w_x}{2}\right)_+, \left(|\Delta_{i,2}| - \tfrac{w_y}{2}\right)_+ \right) \right\|_2, \qquad d_i^{\max} = \left\| \left(|\Delta_{i,1}| + \tfrac{w_x}{2}, |\Delta_{i,2}| + \tfrac{w_y}{2}\right) \right\|_2. \tag{38}$$

For $b = 0$ *both bounds equal* $\mu f_N \|a\|_2$*.*

Proof. Since only the taxel forces are measured, a measure is compatible if and only if its part in each taxel P_i has mass F_i, and these parts are free. The objective $\mu \sum_i \int_{P_i} g(\cdot; q)\, d\nu_N$ is a sum of one term per taxel, so each term is optimized on its own. For one taxel, $F_i \inf_{P_i} g \leq \int_{P_i} g\, d\nu_N \leq F_i \sup_{P_i} g$, and point loads at points of P_i approach both ends. This gives (37). The function $g(\cdot; q)$ is continuous and convex, so its infimum over P_i is its minimum over the closed rectangle, and its supremum is the maximum over the rectangle, attained at a corner. For $b \neq 0$, $g(x; q) = |b|\, \|x - p\|_2$: the closed forms are the distance from p to the rectangle, and to its farthest corner. □

7 $\mathcal{M}_T$ case (taxel forces and CoP)

Known: the taxel forces F_i and the first moment m_1, i.e., the CoP. **Unknown:** the distribution of $\nu_N^\star$ inside each taxel. The taxel constraints act on each taxel separately, but the first-moment constraint $\int_P x\, d\nu_N = m_1$ couples the taxels.

Theorem 7.1 (Exact lower bound for combined measurements)**.** *On a taxel grid, given* $y_T = (f_N, m_1, F_1, \dots, F_r) = \mathcal{M}_T(\nu_N^\star)$*, for every* $q = (a, b)$*,*

$$\boxed{\begin{aligned} \underline{h}(q; y_T, \mathcal{M}_T) = \mu \min_{u_1,\dots,u_r \in \mathbb{R}^2} \quad & \sum_{i=1}^{r} \|F_i a + bJu_i\|_2 \\ \textit{subject to} \quad & F_i\, \mathrm{lo}_i \leq u_i \leq F_i\, \mathrm{hi}_i, \qquad i = 1, \dots, r, \\ & \sum_{i=1}^{r} u_i = m_1. \end{aligned}} \tag{39}$$

The variables are the local first moments (inequalities componentwise): for $F_i > 0$*,* $u_i = F_i c_i$ *where* c_i *is the local CoP of taxel* i*, constrained to lie in the taxel, and the objective is* $\mu \sum_i F_i\, g(c_i; q)$*, while* $F_i = 0$ *forces* $u_i = 0$*. The problem is an SOCP.*

Proof. ($\geq$) For a compatible ν_N, the local first moments $u_i = \int_{P_i} x\, d\nu_N$ are feasible, and $\int_{P_i} g\, d\nu_N \geq \|F_i a + bJu_i\|_2$ by Lemma 3.12(iii) applied to the restriction of ν_N to P_i. So $h_{\mathcal{X}(\nu_N)}(q)$ is at least the minimum. ($\leq$) Let u be optimal. The point loads $\sum_{i:F_i>0} F_i \delta_{u_i/F_i}$ have the taxel forces F_i, the first moment m_1, and the optimal value. But a point u_i/F_i may lie on an upper edge of its taxel, which the half-open P_i does not contain. We move every point of taxel i slightly toward the local CoP p_i of $\nu_N^\star$ in P_i, by $x \mapsto (1-\varepsilon)x + \varepsilon p_i$. The new point lies in P_i, since $p_i \in P_i$. The taxel forces are unchanged, and so is the first moment, because $\sum_i F_i p_i = m_1$. So the new measure is compatible, and by Lemma 3.12(ii) its value is within $\varepsilon\mu|b|D\sum_i F_i$ of the optimum. Let $\varepsilon \to 0$. The feasible set is compact and nonempty (it contains the local first moments of $\nu_N^\star$), and the objective is convex and continuous; epigraph variables give an SOCP. □

Theorem 7.2 (Exact upper bound for combined measurements)**.** *Under the assumptions of Theorem 7.1, for every $q = (a, b)$,*

$$\begin{aligned} \overline{h}(q; y_T, \mathcal{M}_T) = \mu \max_{\lambda_{ik} \geq 0} \quad & \sum_{i=1}^{r} \sum_{k=1}^{4} \lambda_{ik} \left\| a + bJv_{ik} \right\|_2 \\ \textit{subject to} \quad & \sum_{k=1}^{4} \lambda_{ik} = F_i, \qquad i = 1, \ldots, r, \\ & \sum_{i=1}^{r} \sum_{k=1}^{4} \lambda_{ik} v_{ik} = m_1. \end{aligned} \tag{40}$$

This is a linear program, and an optimal solution exists with at most $r + 2$ nonzero weights.

Proof. ($\leq$) For a compatible ν_N, Lemma 5.5 gives corner weights $\lambda_{ik} \geq 0$ with $\sum_k \lambda_{ik} = F_i$ and $\sum_{i,k} \lambda_{ik} v_{ik} = \sum_i u_i = m_1$. They are feasible, and $h_{\mathcal{X}(\nu_N)}(q) \leq \mu \sum_{i,k} \lambda_{ik} g(v_{ik}; q)$, the objective of λ. ($\geq$) Let λ be feasible. The corner loads $\sum_{i,k} \lambda_{ik} \delta_{v_{ik}}$ have the taxel forces F_i, the first moment m_1, and the objective value, but a corner may lie on an edge owned by a neighboring taxel. As in the proof of Theorem 7.1, moving every point of taxel i by $x \mapsto (1-\varepsilon)x + \varepsilon p_i$, with p_i the local CoP of $\nu_N^\star$ in P_i, gives a compatible measure whose value is within $\varepsilon\mu|b|D\sum_i F_i$ of the objective. Let $\varepsilon \to 0$. The LP is feasible (the corner weights of $\nu_N^\star$ are) and bounded, and a basic optimal solution has at most as many nonzero entries as equality constraints, i.e., $r + 2$. □

Remark 7.3 (Recovering the coarse and taxel-force cases)**.** *For a single taxel ($k_x = k_y = 1$), the constraint of (39) forces $u_1 = m_1$, and Theorem 7.1 reduces to Theorem 4.2; Theorem 7.2 reduces to Corollary 4.5 for the rectangle P, with $r + 2 = 3$ nonzero weights. Dropping the first-moment constraint in (39) and (40) decouples the taxels and gives the $\mathcal{M}_F$ bounds of Theorem 6.1. Consistently with Theorem 3.15 and $\mathcal{M}_T \succeq \mathcal{M}_F$, the $\mathcal{M}_T$ bounds lie between the $\mathcal{M}_F$ bounds:*

$$\underline{h}(q; y_F, \mathcal{M}_F) \leq \underline{h}(q; y_T, \mathcal{M}_T) \leq \overline{h}(q; y_T, \mathcal{M}_T) \leq \overline{h}(q; y_F, \mathcal{M}_F). \tag{41}$$

The difference is the value of measuring the CoP in addition to the taxel forces.

8 Comparison of measurements and tightening under refinement

All results of this section follow from Theorem 3.15, the information chains of Proposition 5.4, and the closed forms of Theorem 6.1.

Proposition 8.1 (Comparison of coarse, taxel and combined measurements)**.** *For every twist* $q = (a, b) \in \mathbb{R}^2 \times \mathbb{R}$*,*

$$\boxed{\begin{aligned}\max\{\underline{h}(q; y_0, \mathcal{M}_0),\, \underline{h}(q; y_F, \mathcal{M}_F)\} \le \underline{h}(q; y_T, \mathcal{M}_T) \le h_{\mathcal{X}(\nu_N^\star)}(q),\\ h_{\mathcal{X}(\nu_N^\star)}(q) \le \overline{h}(q; y_T, \mathcal{M}_T) \le \min\left\{\overline{h}(q; y_0, \mathcal{M}_0),\, \overline{h}(q; y_F, \mathcal{M}_F)\right\},\end{aligned}} \tag{42}$$

with $\underline{h}(q; y_0, \mathcal{M}_0) = \mu\|f_N a + bJm_1\|_2$*. The bounds of* $\mathcal{M}_0$ *and* $\mathcal{M}_F$ *are not ordered in general, in either direction.*

Proof. Since $\mathcal{M}_T = \mathcal{M}_0 \vee \mathcal{M}_F$, $\mathcal{M}_T$ is at least as informative as both (Proposition 3.14(iii)), and Theorem 3.15 gives the chain. The bounds of $\mathcal{M}_0$ and $\mathcal{M}_F$ are not ordered, in either direction. With a single taxel, $\mathcal{M}_F$ measures only f_N, so $\mathcal{M}_0$ is tighter, e.g. $\underline{h}(q; y_F, \mathcal{M}_F) = \mu f_N |b| \operatorname{dist}(p, P) < \underline{h}(q; y_0, \mathcal{M}_0) = \mu f_N |b| \|c - p\|_2$ when the CoP c is not the point of P closest to p. With fine taxels, $\mathcal{M}_F$ is tighter: its bounds converge to the truth (Proposition 8.3), while those of $\mathcal{M}_0$ do not in general (Example 4.7). □

Proposition 8.2 (Tightening under taxel refinement)**.** *Let* $\mathcal{M}'_F$, $\mathcal{M}'_T$ *and* $\mathcal{M}_F$, $\mathcal{M}_T$ *be the taxel-force and combined measurements of the grids* $G(k'_x, k'_y)$ *and* $G(k_x, k_y)$*, with* $k_x \mid k'_x$ *and* $k_y \mid k'_y$ *(e.g. dyadic grids, each doubling the previous one). Then, for every* q *and* $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$*, with* $y = \mathcal{M}(\nu_N^\star)$ *and* $y' = \mathcal{M}'(\nu_N^\star)$*,*

$$\boxed{\underline{h}(q; y, \mathcal{M}) \le \underline{h}(q; y', \mathcal{M}') \le h_{\mathcal{X}(\nu_N^\star)}(q) \le \overline{h}(q; y', \mathcal{M}') \le \overline{h}(q; y, \mathcal{M}),} \tag{43}$$

and the guaranteed sets grow and the possible sets shrink.

Proof. Proposition 5.4(i) and Theorem 3.15. □

The monotonicity is associated with **nested grids**, not with the number of taxels r: a 3×3 and a 4×4 grid carry different information (Proposition 5.4(i)), and nothing guarantees that the second gives tighter bounds. The gap bounds below, on the other hand, only require small taxel diameters.

Proposition 8.3 (Gap bound)**.** *For every twist* $q = (a, b)$ *and* $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$ *on a grid of taxel diameter* D*,*

$$\boxed{0 \le \overline{h}(q; y, \mathcal{M}) - \underline{h}(q; y, \mathcal{M}) \le \mu|b|\, f_N\, D.} \tag{44}$$

Hence, along any sequence of grids with $D \to 0$*, the four bounds of* $\mathcal{M}_F$ *and* $\mathcal{M}_T$ *converge to* $h_{\mathcal{X}(\nu_N^\star)}(q)$*.*

Proof. By Theorem 6.1, the $\mathcal{M}_F$ gap is $\mu \sum_i F_i \left(\sup_{P_i} g - \inf_{P_i} g\right)$. The function $g(\cdot; q)$ is $|b|$-Lipschitz (Lemma 3.12(ii)), so each bracket is at most $|b|D$ and the sum is at most $\mu|b| f_N D$. The $\mathcal{M}_T$ bounds lie inside the $\mathcal{M}_F$ bounds (41), so their gap is smaller. Convergence follows since $h_{\mathcal{X}(\nu_N^\star)}(q)$ lies between the bounds. □

With the torque taken about the CoP of each candidate measure, the corresponding bound for $\mathcal{M}_F$ had an additional factor 2, which came from the unknown reference point.

The next result answers how much one refinement step tightens the bounds. For a twist with $b \neq 0$, let $p = Ja/b$ be its instantaneous center of rotation and $\hat{r}(x) = (x - p)/\|x - p\|_2$ the unit radial direction at $x \neq p$, with components $\hat{r}_1, \hat{r}_2$.

Theorem 8.4 (First-order rate of tightening)**.** *Consider a taxel grid $G(k_x, k_y)$ (Definition 5.1), with taxels of size $w_x \times w_y$ and diameter $D = \sqrt{w_x^2 + w_y^2}$, and let $\nu_N^\star$ have a density $\rho_N \le \bar\rho$ with respect to area. Then, for every twist $q = (a, b)$ with $b \neq 0$,*

$$\overline{h}(q; y_F, \mathcal{M}_F) - \underline{h}(q; y_F, \mathcal{M}_F) = \mu|b| \int_P \left(|\hat r_1(x)|\, w_x + |\hat r_2(x)|\, w_y\right) d\nu_N^\star(x) + R, \tag{45}$$

$$|R| \le C\, \mu|b|\, \bar\rho\, D^2 \left(\sqrt{|P|} + D\right), \tag{46}$$

where $|P|$ is the area of P and C is an absolute constant ($C = 8$ works). In particular, for square cells of side h,

$$\overline{h}(q; y_F, \mathcal{M}_F) - \underline{h}(q; y_F, \mathcal{M}_F) = \mu|b|\, C_\nu(p)\, h + O(h^2), \tag{47}$$

$$C_\nu(p) \triangleq \int_P \left(|\hat r_1| + |\hat r_2|\right) d\nu_N^\star \in \left[f_N, \sqrt{2} f_N\right], \tag{48}$$

so that, if $f_N > 0$, doubling k_x and k_y divides the $\mathcal{M}_F$ gap by $2 + O(h)$. Since the $\mathcal{M}_T$ gap is at most the $\mathcal{M}_F$ gap (41), the same expression also bounds it from above; a matching lower bound is not proven.

Proof. By Theorem 6.1, the gap is $\mu|b| \sum_{i=1}^r F_i \theta_i$, where the sum runs over the taxels P_i (the constant C of the statement plays no role here), $F_i = \nu_N^\star(P_i)$, and θ_i is the range of $d(x) = \|x - p\|_2$ over P_i. The idea is that, away from p, the distance d is almost linear on a taxel, and the range of a linear function over a rectangle is explicit. Let $\Delta_i = m_i - p$ and $\Lambda(x) \triangleq |\hat r_1(x)| w_x + |\hat r_2(x)| w_y$; then $\Lambda(m_i)$ is the range of $x \mapsto \langle \hat r(m_i), x\rangle$ over P_i.

Far taxels ($\|\Delta_i\|_2 \ge D$). For $x \in P_i$, $d(x) = \langle \hat r(m_i), x - p\rangle + \varepsilon_i(x)$ with $0 \le \varepsilon_i(x) \le D^2/(4\|\Delta_i\|_2)$: the component of $x - p$ orthogonal to $\hat r(m_i)$ is that of $x - m_i$, of norm at most $D/2$. So $|\theta_i - \Lambda(m_i)| \le D^2/(4\|\Delta_i\|_2)$. The direction also varies little on the taxel, $\|\hat r(x) - \hat r(m_i)\|_2 \le D/\|\Delta_i\|_2$, so $|\Lambda(m_i) - \Lambda(x)| \le D^2/\|\Delta_i\|_2$. Since $\|\Delta_i\|_2 \ge \frac{2}{3}\|x - p\|_2$, this gives $|F_i\theta_i - \int_{P_i} \Lambda\, d\nu_N^\star| \le \frac{15}{8} D^2 \int_{P_i} \frac{d\nu_N^\star(x)}{\|x-p\|_2}$.

Near taxels ($\|\Delta_i\|_2 < D$). Both θ_i and $\Lambda(x)$ lie in $[0, D]$, so such a taxel contributes an error of at most DF_i. These taxels lie in the disk $B(p, \frac{3}{2}D)$, whose mass is at most $\frac{9\pi}{4}\bar\rho D^2$.

Sum. With $\rho_N \le \bar\rho$ and $\int_P \frac{dA}{\|x-p\|_2} \le 2\sqrt{\pi|P|}$ (the maximum over sets of area $|P|$ is reached by the disk centered at p), the total error is at most $\mu|b|\bar\rho D^2 \left(\frac{15}{4}\sqrt{\pi|P|} + \frac{9\pi}{4} D\right)$, which gives (45) with $C = 8$. For square cells, $\Lambda(x) = h(|\hat r_1| + |\hat r_2|)$ with $1 \le |\hat r_1| + |\hat r_2| \le \sqrt{2}$ for a unit vector. The ratio of two consecutive gaps is then $\frac{C_\nu(p)h/2 + O(h^2)}{C_\nu(p)h + O(h^2)}$, with $C_\nu(p) \ge f_N > 0$. □

Remark 8.5 (Reading the rate)**.**

- *In terms of the diameter $D = \sqrt{2}h$ of square cells, the $\mathcal{M}_F$ gap is $\frac{C_\nu(p)}{\sqrt{2} f_N}\, \mu|b| f_N D + O(D^2)$, between $\frac{1}{\sqrt{2}}$ and 1 times the bound of Proposition 8.3: that bound is sharp up to a factor at most $\sqrt{2}$. The constant $C_\nu(p)$ depends on how the load is distributed around the center of rotation, and is largest when the load is seen from p along the diagonals of the grid.*
- *Refining in one direction only, $G(k_x, k_y) \to G(2k_x, k_y)$, halves the term $\int |\hat r_1|\, d\nu\, w_x$ only: the gain depends on the direction of the load seen from p.*
- *The bounded density excludes point loads; for measures with concentrated loads near the cell boundaries, one refinement step can leave the gap almost unchanged at coarse resolution.*
- *The same first-order term bounds the $\mathcal{M}_T$ gap from above, by (41). How much the first-moment constraint reduces the gap at high resolution, i.e., the value of measuring the CoP in addition to the taxel forces, is left open here and quantified numerically in the experimental report.*

9 Certified load factor

A controller does not need the bounds direction by direction. It needs to know whether a given load w can be carried, whether it certainly cannot, and by how much the contact should be strengthened (e.g., by squeezing harder). This is a question about the ray $\{sw : s \geq 0\}$: how far along it the guaranteed and the possible sets extend. This section shows that both answers are convex programs for the taxel measurements, without constructing the sets.

Reference point of the load. The load acting on a grasped object is known about a point o fixed to the object or to the hand. By Lemma 3.4, $sw \in \mathcal{X}_o(\nu_N)$ if and only if $sT_o^{-1}w \in \mathcal{X}(\nu_N)$, so a load about o can always be moved to the origin. We keep o explicit below, since the programs are written most simply with it; the support functions about o are $h_{\mathcal{X}_o(\nu_N)}(q) = \mu \int_P \|a + bJ(x - o)\|_2 \, d\nu_N$ and we write $\underline{h}_o$, $\overline{h}_o$, $\mathcal{X}_o^{\pm}$ for the corresponding bounds and sets.

Definition 9.1 (Load factors)**.** *For a nonzero load $w = (f, \tau) \in \mathbb{R}^2 \times \mathbb{R}$, expressed about o, the load factor of ν_N is $s_o^\star(\nu_N; w) \triangleq \max\{s \geq 0 : sw \in \mathcal{X}_o(\nu_N)\}$. The certified (lower) and the upper load factors are*

$$\underline{s}(w; y, \mathcal{M}) \triangleq \inf_{\nu_N \in \Pi_{\mathcal{M}}(y)} s_o^\star(\nu_N; w), \tag{49}$$

$$\overline{s}(w; y, \mathcal{M}) \triangleq \max\left\{s \geq 0 : sw \in \operatorname{cl} \mathcal{X}_o^+(y, \mathcal{M})\right\} \geq \sup_{\nu_N \in \Pi_{\mathcal{M}}(y)} s_o^\star(\nu_N; w). \tag{50}$$

The load w is carried by every compatible measure if $\underline{s}(w; y, \mathcal{M}) \geq 1$, and by none if $\overline{s}(w; y, \mathcal{M}) < 1$. Since $\mathcal{X}_o(\lambda\nu_N) = \lambda\mathcal{X}_o(\nu_N)$, a contact whose normal force is scaled by λ has load factors scaled by λ.

Proposition 9.2 (Certified and upper load factors)**.** *Let $w = (f, \tau) \neq 0$ and $o \in \mathbb{R}^2$.*

(i) For every measurement operator satisfying Assumption 3.9,

$$\boxed{\begin{aligned} &\underline{s}(w; y, \mathcal{M}) = \max\left\{s \geq 0 : sw \in \mathcal{X}_o^-(y, \mathcal{M})\right\} = \inf_{q:\, \langle q, w\rangle = 1} \underline{h}_o(q; y, \mathcal{M}), \\ &\overline{s}(w; y, \mathcal{M}) = \inf_{q:\, \langle q, w\rangle = 1} \overline{h}_o(q; y, \mathcal{M}). \end{aligned}} \tag{51}$$

(ii) On a taxel grid (Definition 5.1), $\underline{s}$ is the infimum of $\mu \sum_i F_i \|a + bJ(c_i - o)\|_2$ over the twists with $a^\top f + b\tau = 1$ and the local CoPs c_i of the taxels ($\mathrm{lo}_i \leq c_i \leq \mathrm{hi}_i$, and $\sum_i F_i c_i = m_1$ for $\mathcal{M}_T$). This is bilinear in (b, c_i), but it becomes convex after the substitution $u_i = bc_i$: for $\sigma \in \{-1, +1\}$ let

$$\boxed{\begin{aligned} V_\sigma \triangleq \mu \inf_{a \in \mathbb{R}^2,\, b \in \mathbb{R},\, u_i \in \mathbb{R}^2} \quad & \sum_{i=1}^{r} F_i \, \|a + J(u_i - b\, o)\|_2 \\ \text{subject to} \quad & a^\top f + b\tau = 1, \qquad \sigma b \geq 0, \\ & \sigma\,(u_i - b\,\mathrm{lo}_i) \geq 0, \quad \sigma\,(b\,\mathrm{hi}_i - u_i) \geq 0, \qquad i = 1, \ldots, r, \\ & \textstyle\sum_{i=1}^{r} F_i u_i = b\, m_1 \qquad (\mathcal{M}_T \text{ only}), \end{aligned}} \tag{52}$$

with $V_\sigma = +\infty$ if infeasible. Then $\underline{s}(w; y, \mathcal{M}) = \min\{V_{+1}, V_{-1}\}$ for $\mathcal{M} \in \{\mathcal{M}_F, \mathcal{M}_T\}$, and each V_σ is an SOCP.

(iii) Under the same assumptions, with the taxel corners v_{ik}, the upper load factor is the SOCP

$$\boxed{\begin{aligned}\overline{s}(w;y,\mathcal{M}_F) &= \min_{a,b,t}\ \mu\sum_{i=1}^{r} F_i t_i \quad s.t.\ \ a^\top f + b\tau = 1,\quad t_i \geq \|a + bJ(v_{ik}-o)\|_2 \quad \forall i,k,\\ \overline{s}(w;y,\mathcal{M}_T) &= \min_{a,b,y,z}\ \sum_{i=1}^{r} F_i y_i + z^\top m_1\\ &\qquad s.t.\ \ a^\top f + b\tau = 1,\quad y_i + z^\top v_{ik} \geq \mu\|a + bJ(v_{ik}-o)\|_2 \quad \forall i,k,\end{aligned}} \tag{53}$$

and equivalently the primal SOCP

$$\overline{s}(w;y,\mathcal{M}) = \max_{s,\,\lambda_{ik}\geq 0,\,\rho_{ik}\in\mathbb{R}^2}\ s \quad s.t. \quad \textstyle\sum_k \lambda_{ik} = F_i, \quad \|\rho_{ik}\|_2 \leq \mu\lambda_{ik}, \tag{54}$$

$$\textstyle\sum_{i,k}\rho_{ik} = sf, \quad \sum_{i,k}(v_{ik}-o)\times\rho_{ik} = s\tau, \tag{55}$$

$$\textstyle\sum_{i,k}\lambda_{ik}v_{ik} = m_1 \quad (\mathcal{M}_T\ \textit{only}), \tag{56}$$

whose optimal λ is a most favorable corner load, with friction forces ρ_{ik} at its corners.

(iv) If $(a^\star, b^\star, u^\star)$ is optimal for the smaller of the two programs in (ii), let $c_i = u_i^\star/b^\star$ if $b^\star \neq 0$, and let (c_i) be any point loads with $\mathrm{lo}_i \leq c_i \leq \mathrm{hi}_i$ (and $\sum_i F_i c_i = m_1$ for $\mathcal{M}_T$) otherwise. Then $\hat{\nu}_N = \sum_i F_i\delta_{c_i}$ is a point load per taxel with $s_o^\star(\hat{\nu}_N; w) = \underline{s}(w;y,\mathcal{M})$: the certificate is attained, by a compatible measure when each c_i lies in its own taxel.

(v) (Decision rule.) If $\underline{s} \geq 1$, every compatible normal force measure carries w. If $\overline{s} < 1$, none does: whatever the actual distribution, no admissible friction field balances w, and the contact slips (quasi-statically). If $\underline{s} < 1 \leq \overline{s}$, the measurement does not decide. At $\underline{s} = 1$ the load is carried but at the onset of sliding for the least favorable compatible measure; $\underline{s} > 1$ gives a strict margin. If $\mathcal{M} \succeq \mathcal{M}'$, then $\underline{s}(w;y',\mathcal{M}') \leq \underline{s}(w;y,\mathcal{M}) \leq s_o^\star(\nu_N^\star; w) \leq \overline{s}(w;y,\mathcal{M}) \leq \overline{s}(w;y',\mathcal{M}')$.

(vi) (Several contacts.) Let K contacts act on the same rigid object, with independent measurements y_k of their normal force measures ν_N^k, the same μ, and patches identified with subsets of a common plane (common coordinates x, torques about the common normal). Assume that the wrench components out of this plane are balanced without friction. The load w is then carried if $w \in \mathcal{X}_o(\nu_N^1) \oplus \cdots \oplus \mathcal{X}_o(\nu_N^K)$ (Minkowski sum). The certified and upper load factors of the grasp are

$$\boxed{\underline{s}_{\mathrm{grasp}}(w) = \inf_{q:\,\langle q,w\rangle=1}\ \sum_{k=1}^{K}\underline{h}_o(q;y_k,\mathcal{M}^k), \qquad \overline{s}_{\mathrm{grasp}}(w) = \inf_{q:\,\langle q,w\rangle=1}\ \sum_{k=1}^{K}\overline{h}_o(q;y_k,\mathcal{M}^k),} \tag{57}$$

and (ii)–(iii) hold with the objectives of the K contacts summed, a shared (a,b), and separate variables per contact. Since $\bigoplus_k \mathcal{X}_o^-(y_k) \subseteq \bigcap_{(\nu_N^k)} \bigoplus_k \mathcal{X}_o(\nu_N^k)$, $\underline{s}_{\mathrm{grasp}}$ is at least the load factor of the sum of the guaranteed sets of the individual contacts.

Proof sketch. (i) *Fixed measure.* The set $\mathcal{X}_o(\nu_N)$ is convex, compact and contains 0, so $\{s \geq 0 : sw \in \mathcal{X}_o(\nu_N)\} = [0, s_o^\star(\nu_N; w)]$. A closed convex set is the set of points satisfying $\langle q, \cdot\rangle \leq h(q)$ for all q [4, Section 13], and this is void for $\langle q, w\rangle \leq 0$ since $h \geq 0$. By positive homogeneity, $s_o^\star(\nu_N; w) = \inf_{\langle q,w\rangle>0} h_{\mathcal{X}_o(\nu_N)}(q)/\langle q, w\rangle = \inf_{\langle q,w\rangle=1} h_{\mathcal{X}_o(\nu_N)}(q)$. *Certified factor.* $sw \in \mathcal{X}_o^-(y)$ if and only if $s \leq s_o^\star(\nu_N; w)$ for every compatible ν_N, which is the first equality. Exchanging the infima over ν_N and over q gives the second. *Upper factor.* The same argument applies to the closed convex set $\mathrm{cl}\,\mathcal{X}_o^+(y)$, which contains 0, is bounded by Assumption 3.9, and has support function $\overline{h}_o$ (Proposition 3.10(i)).

(ii) *Point loads.* By (i) and Theorems 6.1 and 7.1, $\underline{s}$ is the infimum of $\mu \sum_i F_i \|a + bJ(c_i - o)\|_2$ over the twists with $a^\top f + b\tau = 1$ and the local CoPs c_i of the taxels. *Convexification.* The objective is bilinear in (b, c_i). Split the feasible set by the sign of b and set $u_i = bc_i$. Then $bJ(c_i - o) = J(u_i - b\,o)$, so the objective is unchanged; the constraints on c_i become linear, $\sigma(u_i - b\,\mathrm{lo}_i) \geq 0$, $\sigma(b\,\mathrm{hi}_i - u_i) \geq 0$ and $\sum_i F_i u_i = b\,m_1$; and for $\sigma b > 0$ the map $c_i \mapsto u_i$ is a bijection. *The case $b = 0$.* The original objective is $\mu f_N \|a\|_2$ for every feasible (c_i), and in the new variables the box constraints force $u_i = 0$, which gives the same value. *SOCP.* The objective is a sum of norms of affine functions and the constraints are linear, so epigraph variables give an SOCP.

(iii) $\mathcal{M}_F$. By (i) and Theorem 6.1, $\overline{s} = \inf \mu \sum_i F_i \max_k \|a + bJ(v_{ik} - o)\|_2$ over $\langle q, w \rangle = 1$, which is the stated program in epigraph form. $\mathcal{M}_T$. The LP of Theorem 7.2 (with $g(v_{ik}; q)$ replaced by $\|a + bJ(v_{ik} - o)\|_2$) is feasible and bounded, so by LP duality $\overline{h}_o(q) = \min\{\sum_i F_i y_i + z^\top m_1 : y_i + z^\top v_{ik} \geq \mu \|a + bJ(v_{ik} - o)\|_2\}$. Minimizing jointly over (q, y, z) gives the stated program, with second-order cone constraints. *Primal form.* For fixed corner weights λ, the wrenches of corner forces ρ_{ik} with $\|\rho_{ik}\|_2 \leq \mu\lambda_{ik}$ form a compact convex set, whose support function in direction q is $\mu \sum_{i,k} \lambda_{ik} \|a + bJ(v_{ik} - o)\|_2$, since $\langle q, (\sum \rho, \sum (v - o) \times \rho) \rangle = \sum_{i,k} \langle a + bJ(v_{ik} - o), \rho_{ik} \rangle$. Maximizing over the feasible λ gives $\overline{h}_o(q)$ by the LP above, so the projection of the feasible set of (λ, ρ) onto the wrench has the support function of $\mathrm{cl}\,\mathcal{X}_o^+(y)$, and equals it.

(iv) When each c_i lies in its own taxel, $\hat{\nu}_N$ is compatible, so $\underline{s} \leq s_o^\star(\hat{\nu}_N; w)$. With $q^\star = (a^\star, b^\star)$, by (i), $s_o^\star(\hat{\nu}_N; w) \leq h_{\mathcal{X}_o(\hat{\nu}_N)}(q^\star) = \mu \sum_i F_i \|a^\star + b^\star J(c_i - o)\|_2 = \underline{s}(w; y, \mathcal{M}) \leq s_o^\star(\hat{\nu}_N; w)$.

(v) Definitions, and Theorem 3.15 for the ordering.

(vi) For fixed measures, $\bigoplus_k \mathcal{X}_o(\nu_N^k)$ is convex, compact, contains 0, and has support function $\sum_k h_{\mathcal{X}_o(\nu_N^k)}$, so the argument of (i) applies. The compatible family is a product of the individual families and the k-th term depends only on ν_N^k, so the infimum (resp. supremum) of the sum over compatible measures is the sum of the infima (resp. suprema). For the inclusion, if $w_k \in \mathcal{X}_o^-(y_k)$ for each k, then $w_k \in \mathcal{X}_o(\nu_N^k)$ for every compatible ν_N^k, so $\sum_k w_k \in \bigoplus_k \mathcal{X}_o(\nu_N^k)$. □

Remark 9.3 (On Proposition 9.2).

- *Why not a grid of directions. $\underline{h}_o(\cdot; y)$ is an infimum of convex functions of q, and is not convex in general. Minimizing it over a finite set of directions gives a value $\geq \underline{s}(w; y, \mathcal{M})$, which may certify a load that is not guaranteed. Minimizing jointly over q and the point loads, after the change of variables $u_i = bc_i$, gives the exact value as a convex program. The upper bound $\overline{h}_o(\cdot; y)$ is convex (a support function), so $\overline{s}$ is a convex minimization from the start.*

- *Coarse case. With a single taxel and $o = c$ ($\mathcal{M}_0$), the constraint forces $c_1 = c$, and (ii) gives $\underline{s} = \mu f_N / \|f\|_2$ if $\tau = 0$ and $\underline{s} = 0$ if $\tau \neq 0$ (take $a = 0$, $b = 1/\tau$). The coarse measurement never certifies a torsional load about the CoP (Corollary 4.3).*

- *Membership tests for general operators. When $\overline{h}$ can be evaluated with a maximizing wrench (a support point of $\mathcal{X}^+$) but no closed form is available, e.g., through moment relaxations of the GMP, the distance from w to $\mathrm{cl}\,\mathcal{X}^+$ can be computed by the Frank–Wolfe method, whose linear minimization oracle is exactly this support point. For the taxel measurements, (iii) is exact and cheaper.*

- *Scope. The certificate is quasi-static and applies to the measured state. In a feedback loop the load and the contact change between decisions, which requires a separate margin. In (vi), the assumption on the out-of-plane components matters: with a free object, offset normal forces at opposing contacts create a couple about an in-plane axis that must be balanced by friction, which the planar model does not account for.*

References

[1] L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, Revised ed., CRC Press, Boca Raton, FL, 2015.

[2] J.-B. Lasserre, *Moments, Positive Polynomials and Their Applications*, Imperial College Press Optimization Series, vol. 1, Imperial College Press, London, 2010.

[3] G. Winkler, "Extreme points of moment sets," *Mathematics of Operations Research*, vol. 13, no. 4, pp. 581–587, 1988.

[4] R. T. Rockafellar, *Convex Analysis*, Princeton University Press, Princeton, NJ, 1970.

[5] S. Alexander and J. H. Maddocks, "Bounds on the friction-dominated motion of a pushed object," *The International Journal of Robotics Research*, vol. 12, no. 3, pp. 231–248, 1993.

[6] S. Goyal, A. Ruina, and J. Papadopoulos, "Planar sliding with dry friction, Part 1," *Wear*, vol. 143, pp. 307–330, 1991.